



How to make nice-looking plots

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LIGO SURF Presentation
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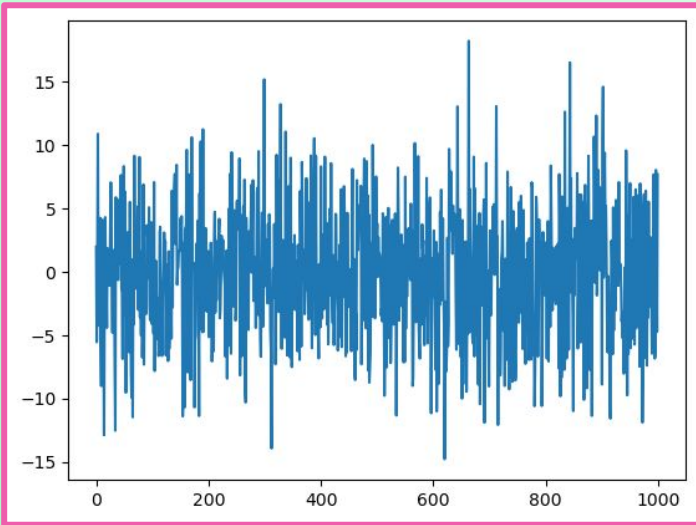
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Simple 1D data

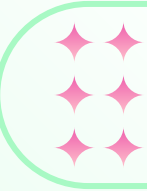
Example plot

```
import numpy as np
import matplotlib.pyplot as plt

data = np.random.normal(loc=0, scale=5, size=1000)
plt.plot(data)
```



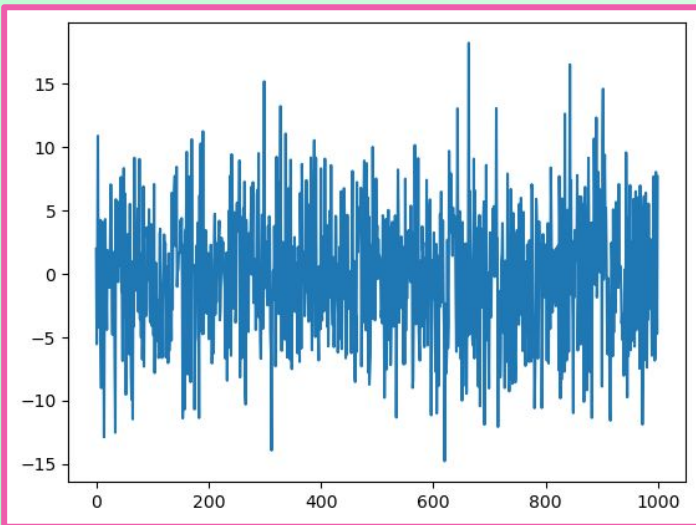
Issues with this plot



Example plot

```
import numpy as np
import matplotlib.pyplot as plt

data = np.random.normal(loc=0, scale=5, size=1000)
plt.plot(data)
```



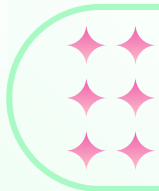
Issues with this plot

Major issues

- No axis labels
- No gridlines
- Too small on screen
- Can't read axes
- Plot not well thought out

Minor issues

- Sans-serif font
- Default colors



My tips for nice plots

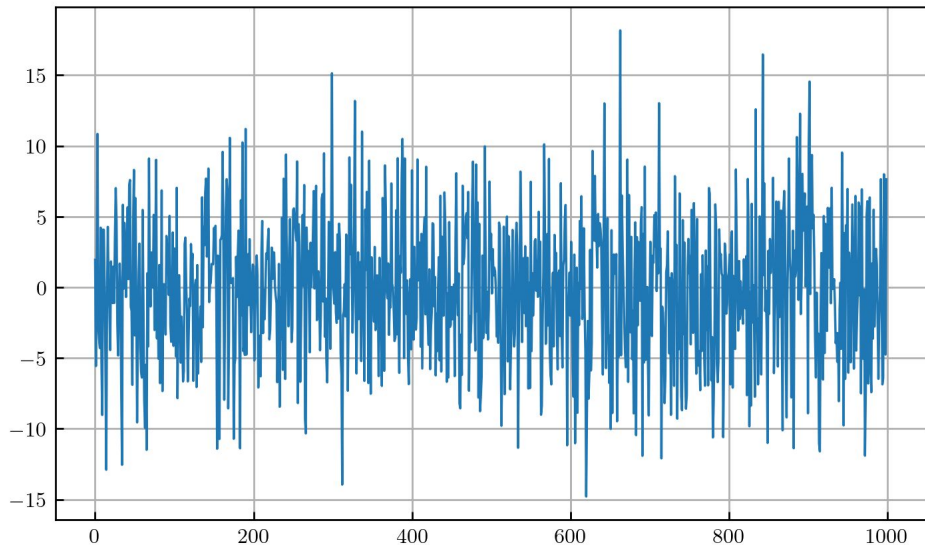
```
import matplotlib
matplotlib.rcParams.update(matplotlib.rcParamsDefault)
fontsize = 9
# ALWAYS USE figsize = (3.375, X) for column plots
# figsize = (6.75, X) for rows
params = {
    'axes.labelsize': fontsize,
    'font.size': fontsize,
    'legend.fontsize': 8,
    'xtick.labelsize': fontsize,
    'ytick.labelsize': fontsize,
    'axes.titlesize': fontsize,
    'lines.linewidth': 1,
    'xtick.direction': 'in',
    'ytick.direction': 'in',
    'text.usetex': True,
    'font.family': 'Serif',
    'font.serif': 'Computer Modern Roman',
    'axes.grid': True,
    'figure.figsize': (6.75, 4),
    'figure.dpi': 250,
}

for param in params.keys():
    matplotlib.rcParams[param] = params[param]
```

Use good defaults

Example plot

```
plt.plot(data)  
plt.show()
```



Issues with this plot

Major issues

- No axis labels
- 😊 No gridlines
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- Default colors

Example plot

Major issues



No axis labels



No gridlines



Too small on screen



Can't read axes



Plot not well thought out

Minor issues



Sans-serif font



Default colors

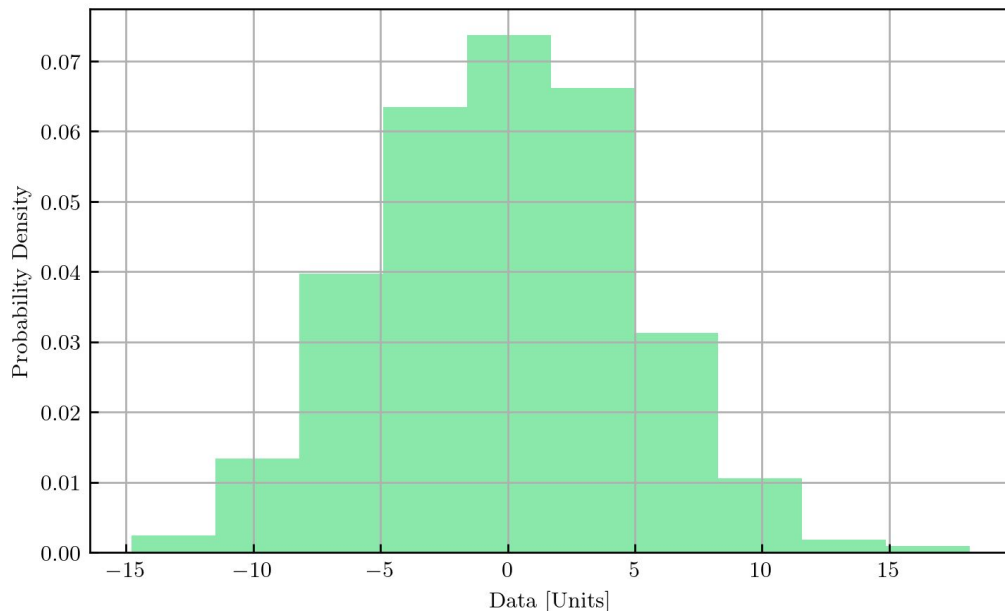
```
fig, ax = plt.subplots()
```

```
ax.hist(data, color='#8ae8aa', density=True)
```

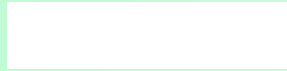
```
ax.set_xlabel('Data [Units]')
```

```
ax.set_ylabel('Probability Density')
```

```
plt.show()
```

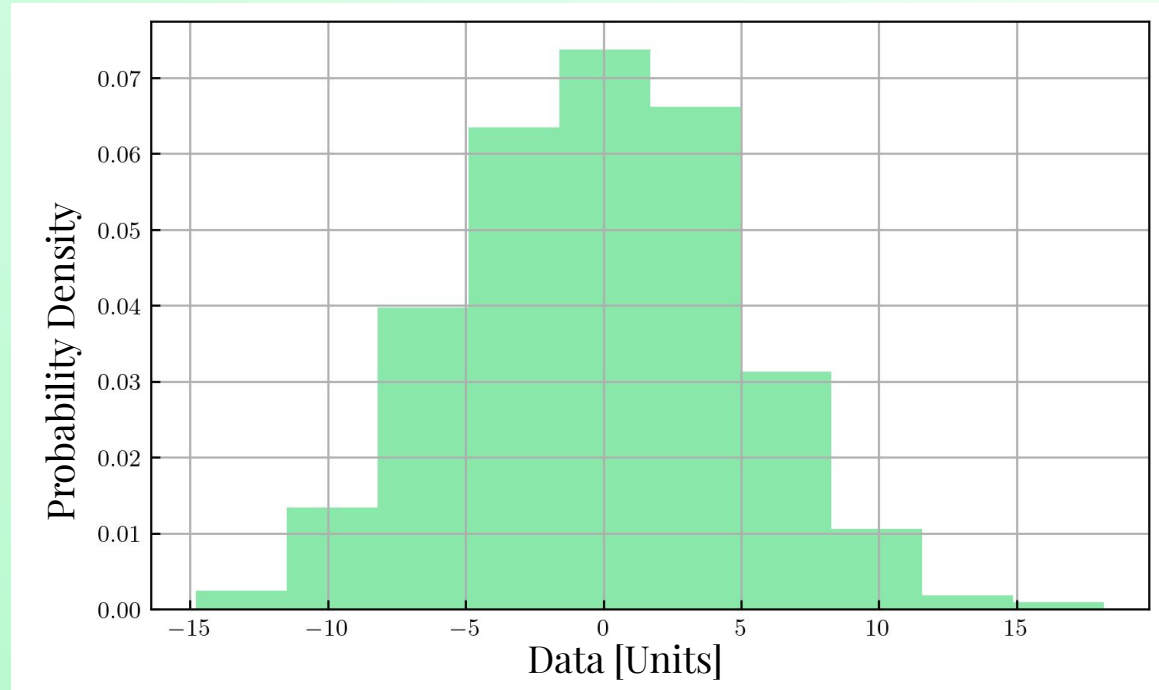
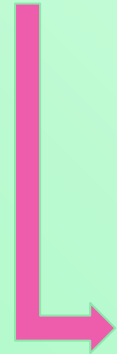


Edit axes directly on slides



+

Data [Units]



Finally

Major issues



No axis labels



No gridlines



Too small on screen



Can't read axes



Plot not well thought out

Minor issues



Sans-serif font

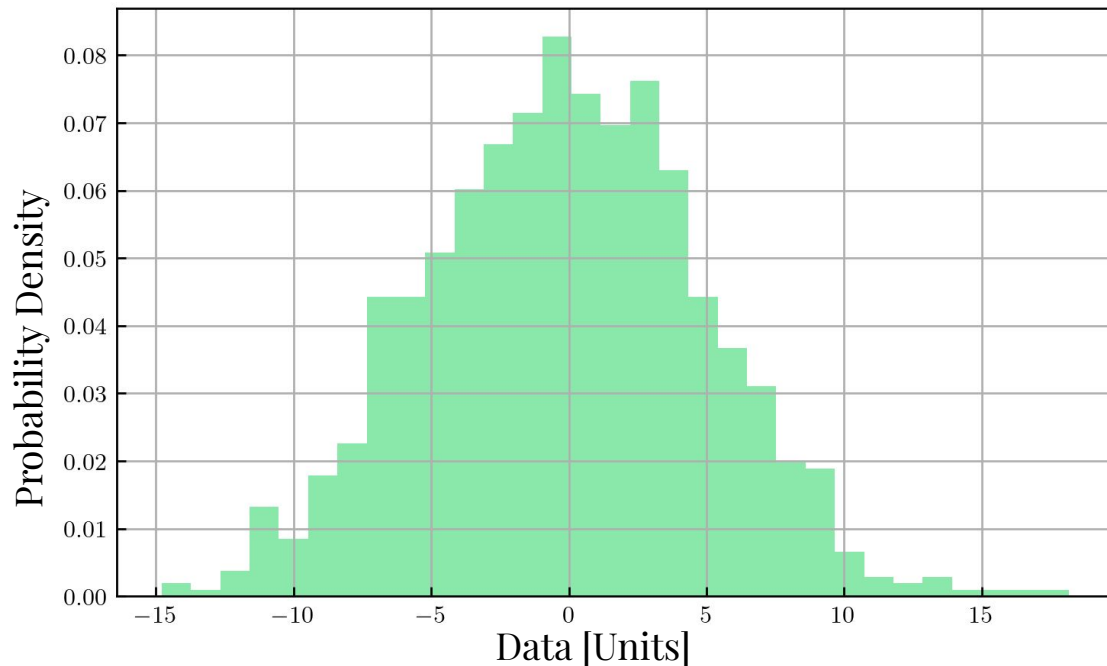


Default colors

```
fig, ax = plt.subplots()

ax.hist(data, color='#8ae8aa', bins = int(np.sqrt(len(data))), density=True)
ax.set_xlabel('Data [Units]')
ax.set_ylabel('Probability Density')

plt.show()
```

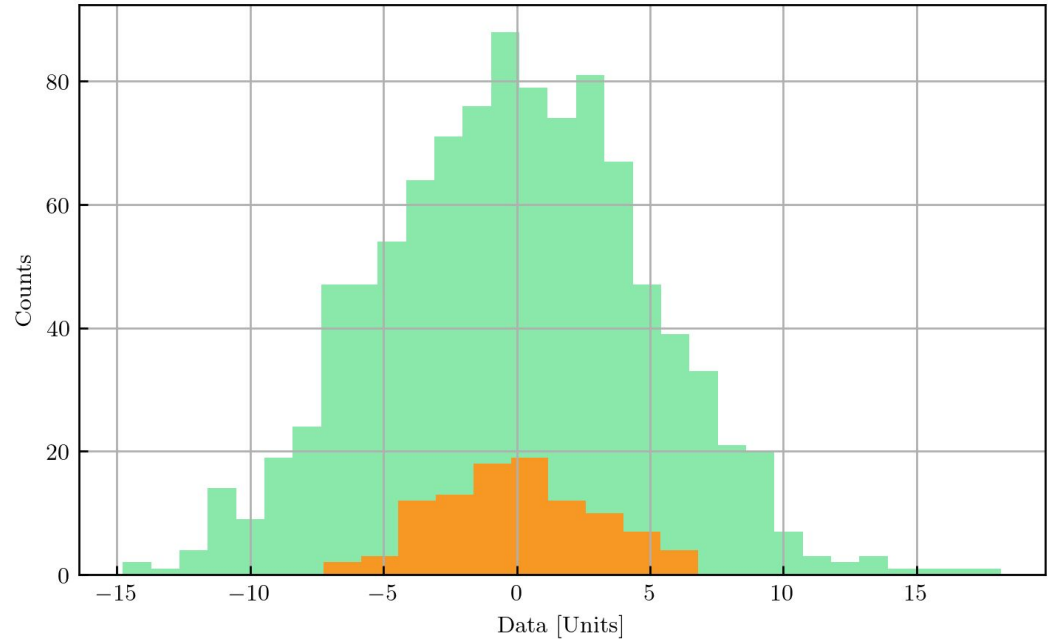


Common Issues

What's the issue with this plot?

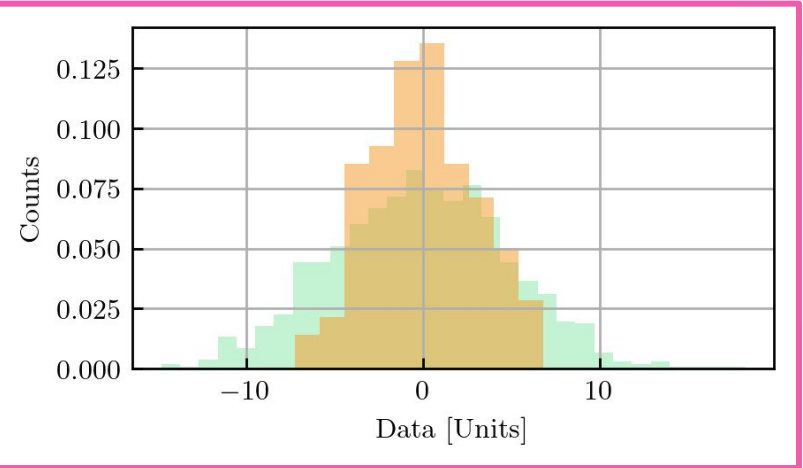
```
data = np.random.normal(loc=0, scale=5, size=1000)  
data_2 = np.random.normal(loc=0, scale=3, size=100)
```

```
fig, ax = plt.subplots()  
ax.hist(data, color='#8ae8aa', bins = int(np.sqrt(len(data))))  
ax.hist(data_2, color='#f79824', bins = int(np.sqrt(len(data_2))))  
  
ax.set_xlabel('Data [Units]')  
ax.set_ylabel('Counts')  
  
plt.show()
```

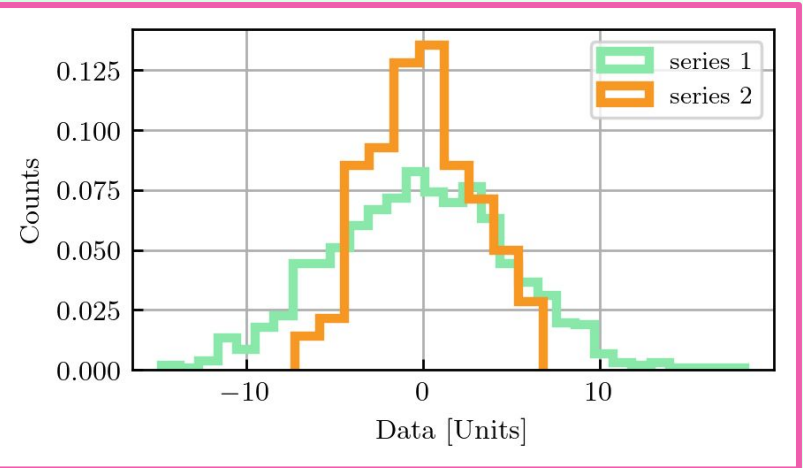


Common Issues

```
fig, ax = plt.subplots(figsize=(3.75, 2))  
  
ax.hist(data, color='#8ae8aa', bins = int(np.sqrt(len(data))),  
        density=True, alpha=0.5)  
ax.hist(data_2, color='#F79824', bins = int(np.sqrt(len(data_2))),  
        density=True, alpha=0.5)  
  
ax.set_xlabel('Data [Units]')  
ax.set_ylabel('Counts')  
  
plt.show()
```



```
fig, ax = plt.subplots(figsize=(3.75, 2))  
ax.hist(data, color='#8ae8aa', bins = int(np.sqrt(len(data))),  
        density=True, histtype='step', lw=3, label='series 1')  
ax.hist(data_2, color='#F79824', bins = int(np.sqrt(len(data_2))),  
        density=True, histtype='step', lw=3, label='series 2')  
  
ax.set_xlabel('Data [Units]')  
ax.set_ylabel('Counts')  
ax.legend()
```



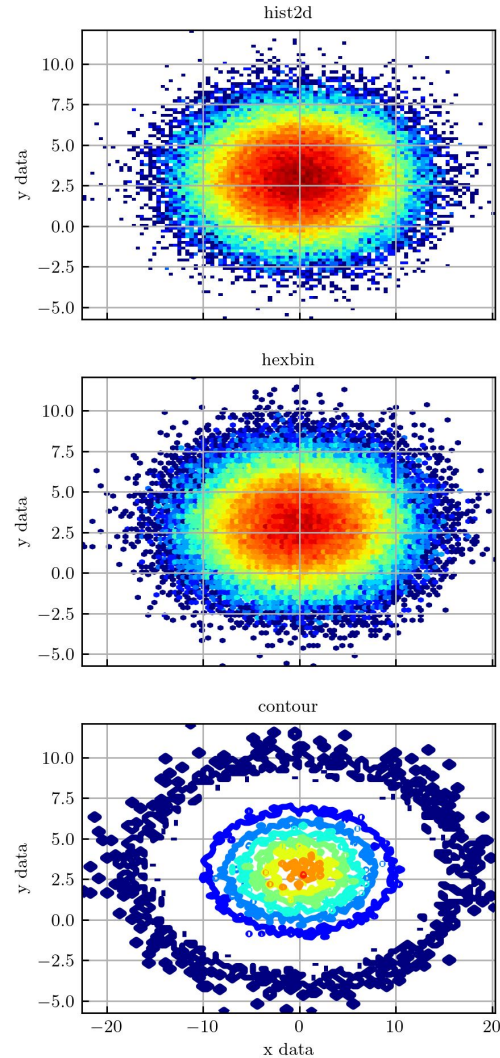


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**Let's up the
dimensions**

Heat map plots

Issues with these plots



Heat map plots

```
import matplotlib.colors as mcolors

data = np.random.normal(loc=0, scale=5, size=100000)
data2 = np.random.normal(loc=3, scale=2, size=100000)

fig, axes = plt.subplots(nrows=3, ncols=1,
                        figsize=(3.75, 9),
                        sharex=True, sharey=True)

axes[0].set_title('hist2d')
axes[0].set_ylabel('y data')

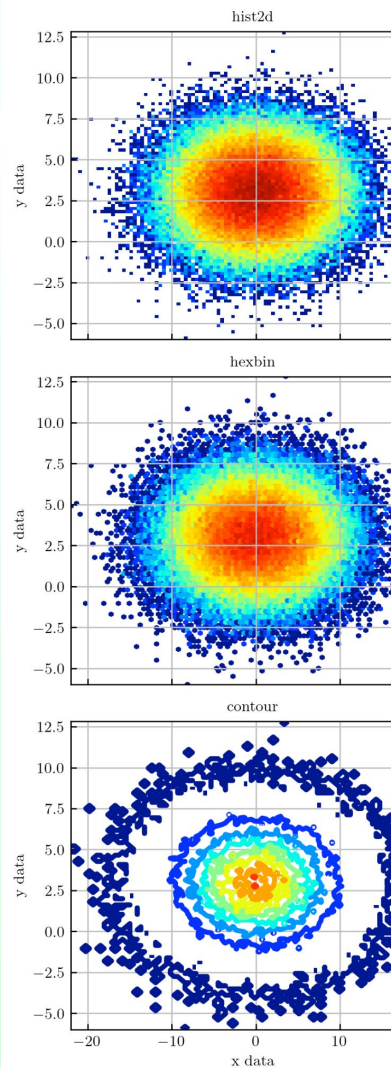
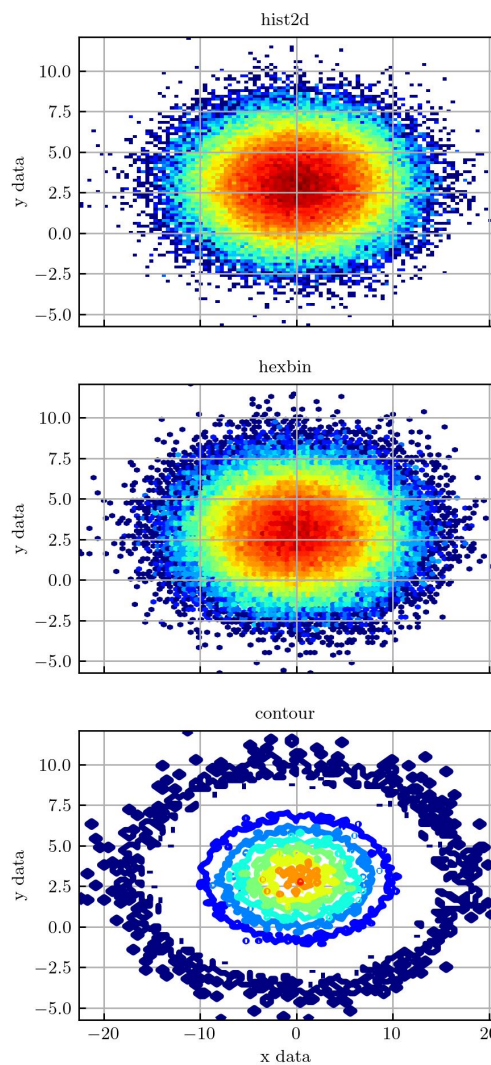
counts, ybins, xbins, image = axes[0].hist2d(data, data2,
                                             bins=100, cmap='jet',
                                             norm=mcolors.LogNorm())

axes[1].set_title('hexbin')
axes[1].hexbin(data, data2, bins=100, cmap='jet', norm=mcolors.LogNorm())
axes[1].set_ylabel('y data')

axes[2].set_title('contour')
axes[2].contour(counts.transpose(),
               extent=[ybins[0], ybins[-1], xbins[0], xbins[-1]],
               linewidths=3,
               cmap='jet')

axes[2].set_xlabel('x data')
axes[2].set_ylabel('y data')

fig.tight_layout()
plt.show()
```



Heat map plots

```
import matplotlib.colors as mcolors

data = np.random.normal(loc=0, scale=5, size=100000)
data2 = np.random.normal(loc=3, scale=2, size=100000)

fig, axes = plt.subplots(nrows=3, ncols=1,
                        figsize=(3.75, 9),
                        sharex=True, sharey=True)

axes[0].set_title('hist2d')
axes[0].set_ylabel('y data')

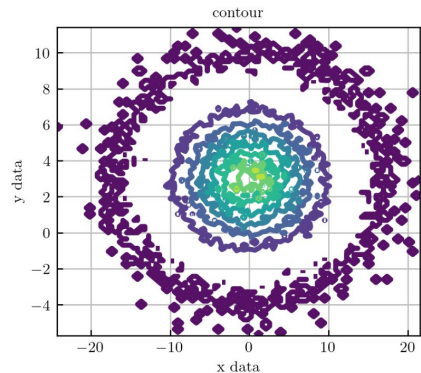
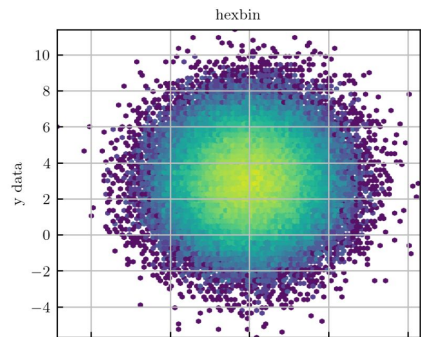
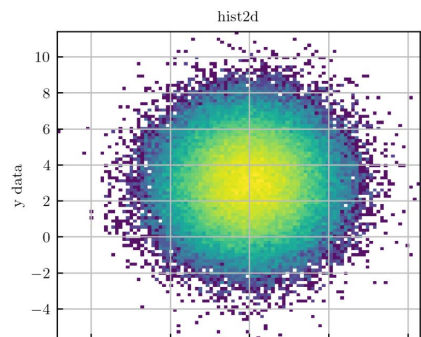
counts, ybins, xbins, image = axes[0].hist2d(data, data2,
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               linewidths=3,
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axes[2].set_xlabel('x data')
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fig.tight_layout()
plt.show()
```



Take home tips

- Use perceptually uniform colormaps
- Make sure your plot is colorblind friendly
- Legible axis labels
- Save to pdf for papers!

Table of contents



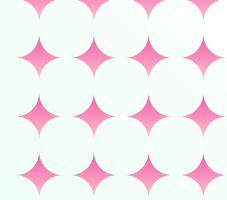
01

Intro to the parent
function

02

Important terms you
need to know





A simple intro to rational functions!

Welcome to the wonderful world of rational functions!

So far, you might have heard of linear, quadratic, or exponential functions, but **rational is a whole new ball game**. Let's get right into it

A rational function is any function that is a **RATIO** of 2 polynomials

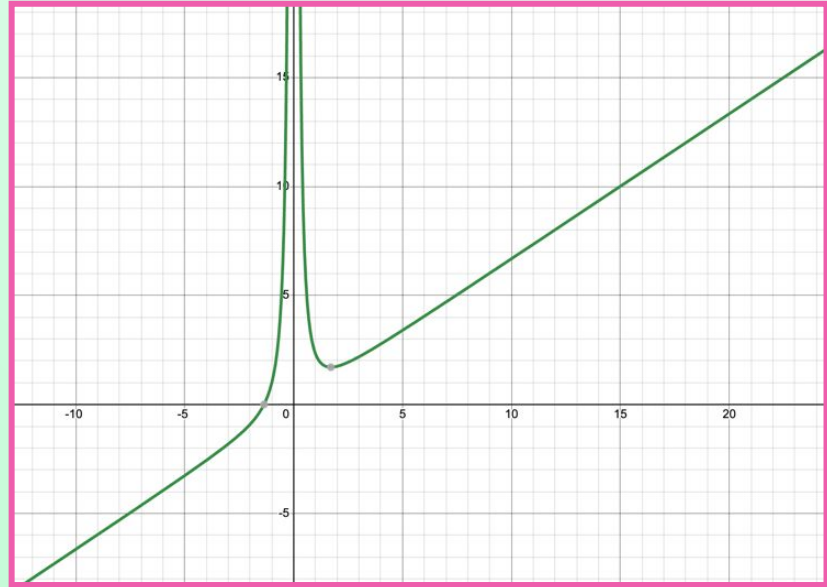
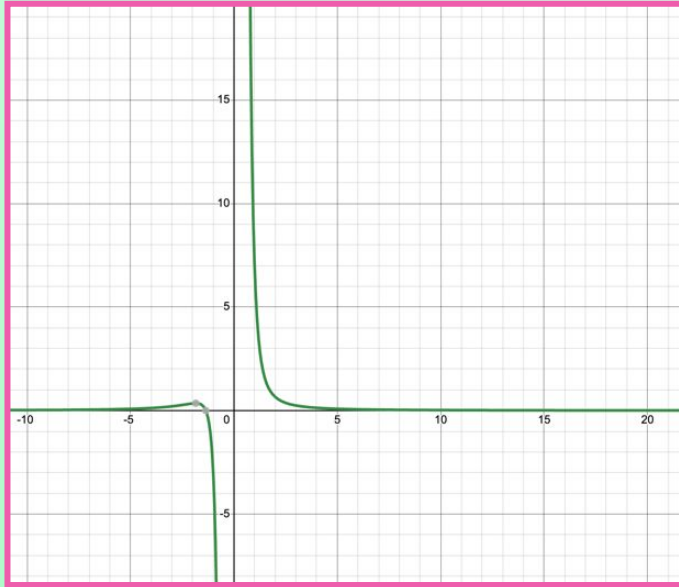
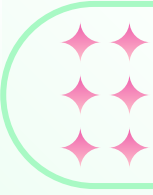
A ratio can be written like a fraction, with a polynomial in the numerator and a polynomial in the denominator

However, the denominator cannot equal zero

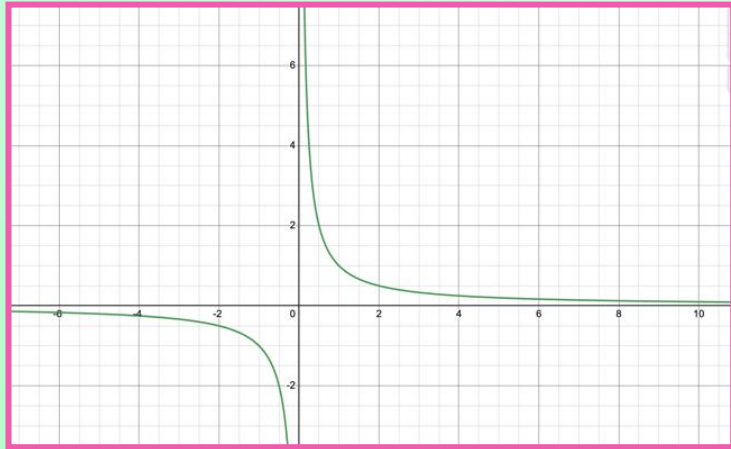


Rational function equations

Here are some examples of rational function equations:



The parent function of a rational function can be written as $y = 1/x$, and this is what it looks like as a table and as a graph



x	$1/x$
-2	-0.5
-1	-1
0	undefined
1	1
2	0.5
3	0.333333
4	0.25

The function is **NOT continuous at $x = 0$** , meaning there is **no (x, y) pair with $x = 0$** for this function

This is because as we mentioned earlier, **dividing by 0 is undefined**

The function is **approaching $x = 0$** , and it is getting closer and closer and closer, but **never quite reaches it**

Why?

Try plugging in:

- x values slightly larger than 0
- Values slightly smaller than 0

See what happens to the y value of your coordinate pair

The values either skyrocket to:

- Positive infinity
- Negative infinity in the opposite case

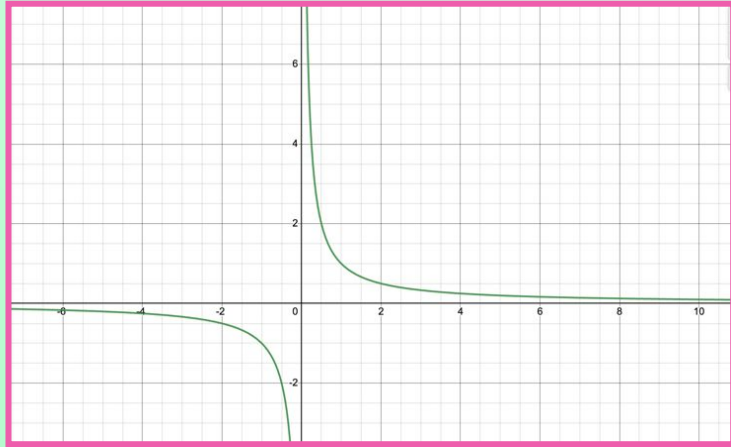
The function **approaches $y = 0$** but **never quite reaches that line**

Why?

- Because the **denominator increases a lot faster than the numerator**
- As the value we plug in for x increases, **the denominator gets larger so the fraction gets smaller and smaller**
- No matter how small our value is, **it will never become zero**
- If you plug in **smaller values for x** , you will end up getting **super large values** because the **numerator is higher now**. Either way you **do not reach $y = 0$**

There's actually a special name for **this phenomenon** happening **with the x and y** values that we cannot reach. This will be covered in the **“Important Terms”** section

Before we move on to “important terms”, the last characteristic of the parent function we will cover is **domain and range of $1/x$**



- We know that **$x = 0$ is not part of the domain** because you can't divide by 0
- The domain is **all x values except for $x = 0$**
- For the range, we covered earlier that **$y = 0$ is not part of the function either**
- So the **range is all y values except for $y = 0$**

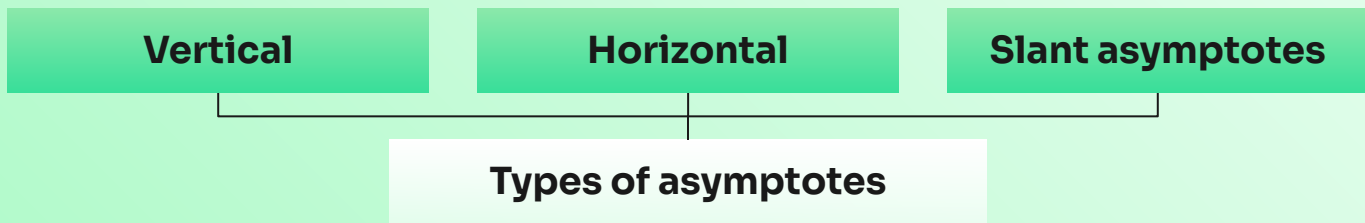


02

Important terms you need to know

Pay very close attention to the upcoming terms

- In section 1, we talked about certain x and y values (**$x = 0$ and $y = 0$**) that our **parent function was never able to reach**. Those are called **asymptotes**
- An asymptote occurs when you try to approach a certain line or curve but never quite reach it, **only get infinitely close**



Vertical

Horizontal

Slant asymptotes

- Occurs at an x value that causes the **denominator to equal 0**
- The **numerator cannot also be zero** for this x value. For our parent function, this was $x = 0$
- If we take another example, $1/(x - 3)$, the vertical asymptote would be $x = 3$, because if you plug 3 into the function, you get 0 for the denominator

Vertical

Horizontal

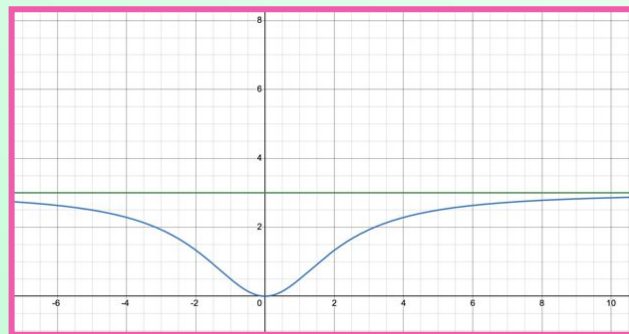
Slant asymptotes

- Occurs at a y value that the function gets close to but never reaches
- If the **denominator polynomial has a higher degree than the numerator**, like $y = x/(x^2 + 5)$, the **horizontal asymptote is ALWAYS 0**

- If the **denominator and numerator have equal degrees**, the **horizontal asymptote is the ratio of the 2 leading coefficients**
- if we have $3x^2/(x^2 + 5)$, the horizontal asymptote would be **$y = 3$** because dividing the leading coefficients would be $3/1 = 3$

— $3x^2/(x^2 + 5)$

— $y = 3$

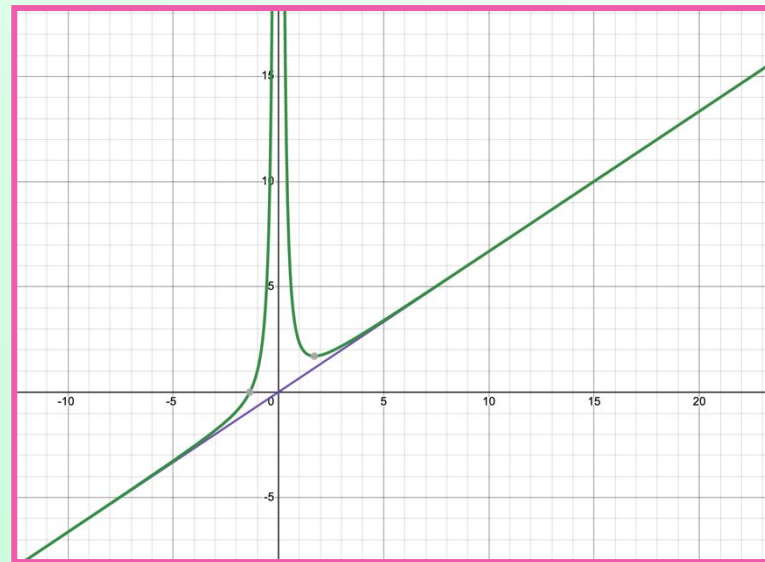


Vertical

Horizontal

Slant asymptotes

- **If the numerator has a higher degree than the denominator**, you have a slant asymptote which is your third type
- You find your slant by **dividing your numerator by your denominator**
- We will go more in depth about this in another lesson



Rational function with a slant asymptote

Thanks

Do you have any questions?

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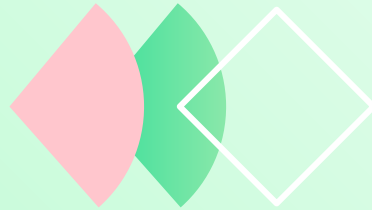
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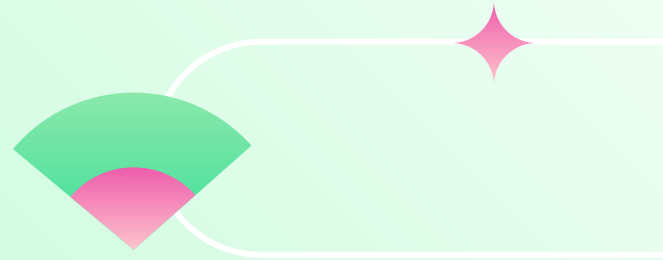


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Pana



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Bro



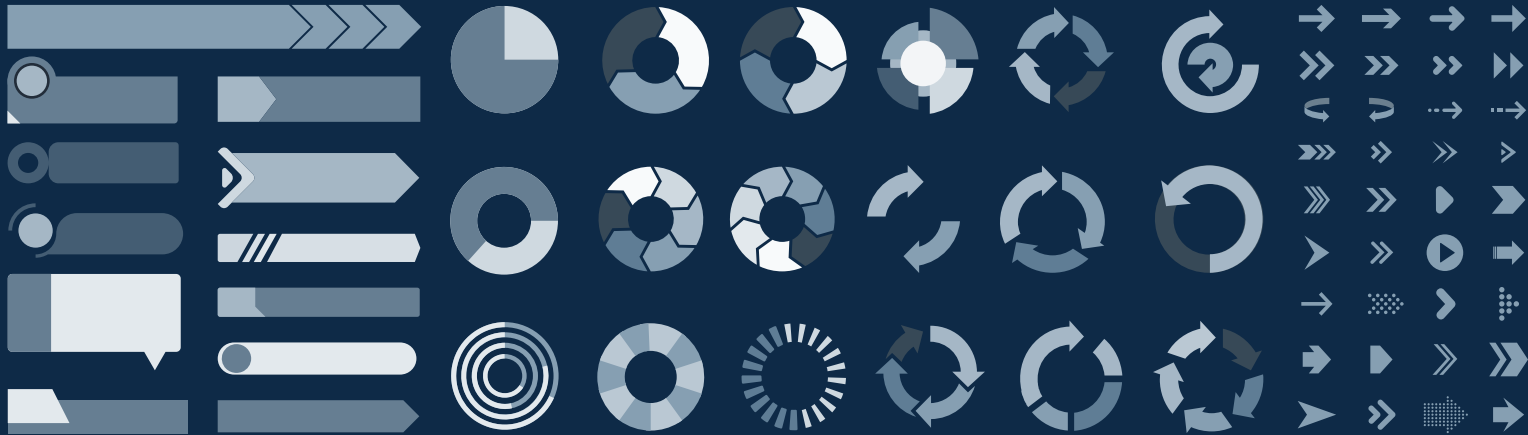
Rafiki



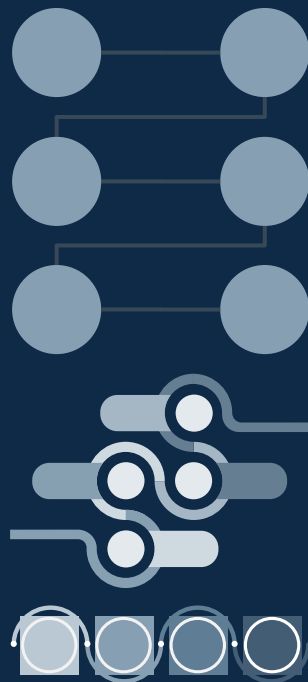
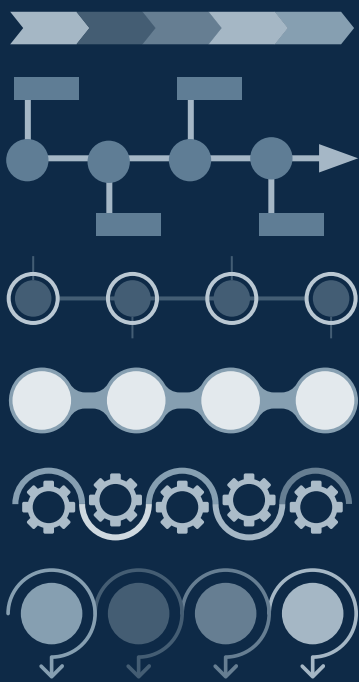
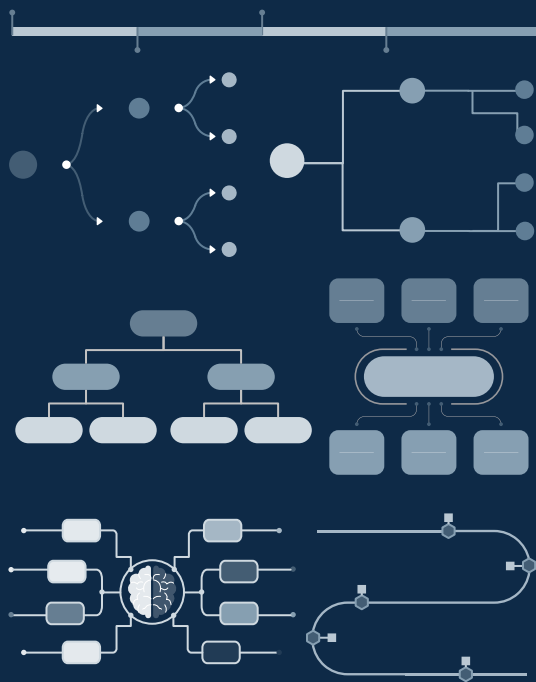
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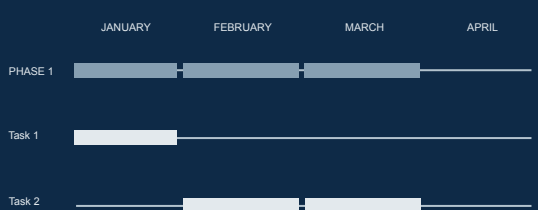
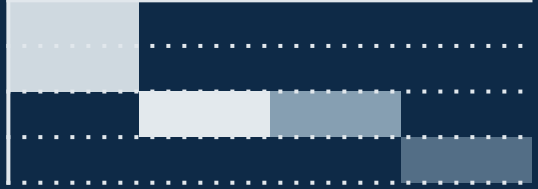
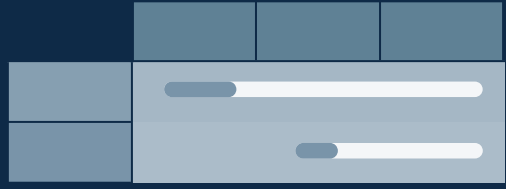
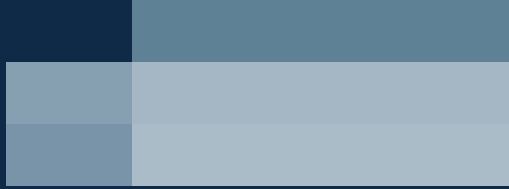
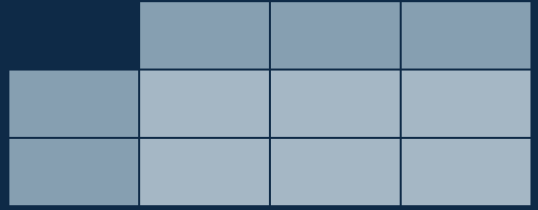
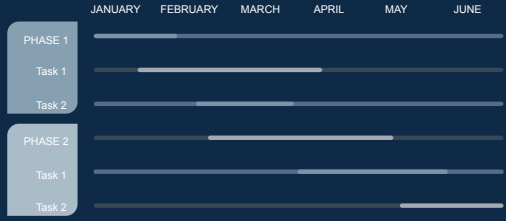
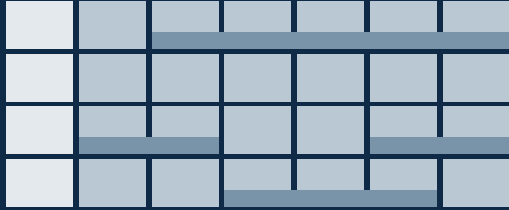
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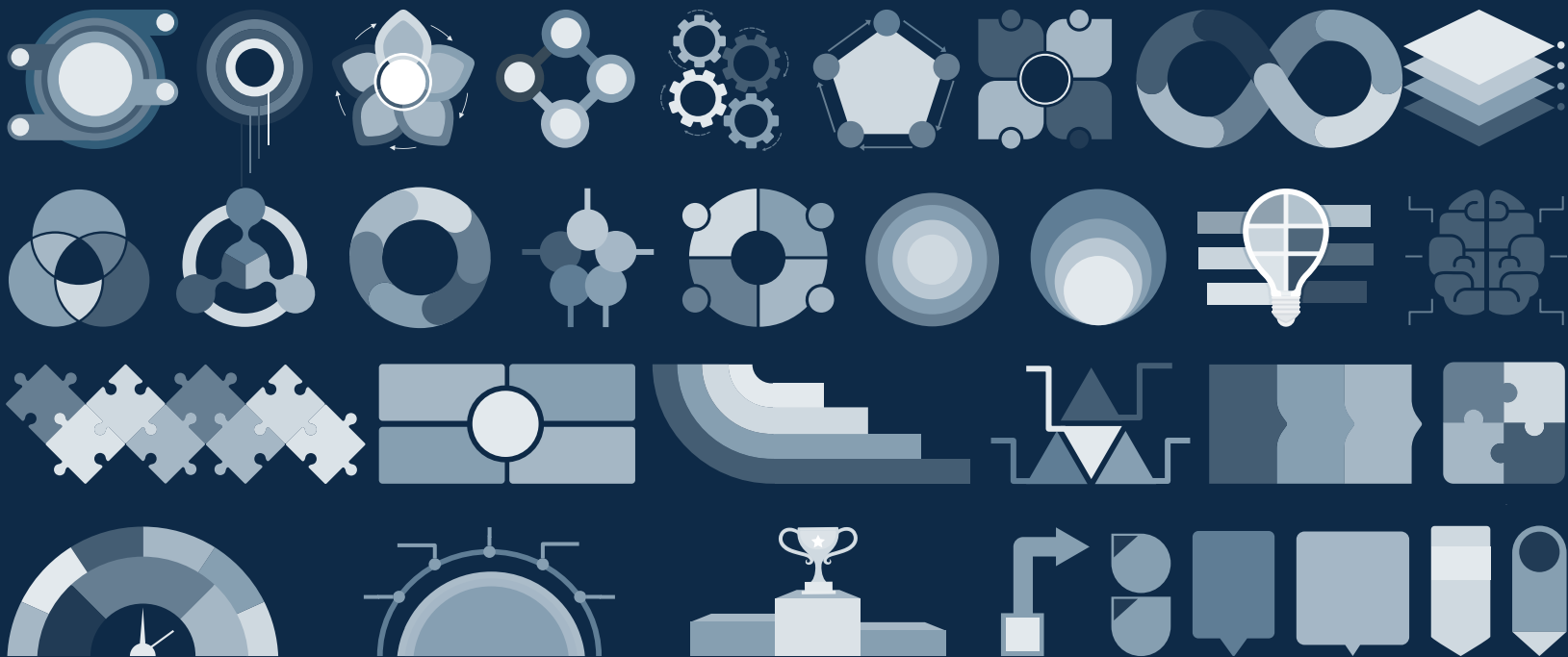
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